

Software Applications for Computer Science

Software Skills for the Real World

Outline

- What do you learn here?
- Why 2026 is special
- Who am I?
- Assessment methods and grading criteria
- What do already know?
- Command Line Interface
- Structure of this seminar
- Practical example: Self deployed change detection for website monitoring

What do you learn here?

- Software skills for the real world
 - Software development is complex
 - Knowing the right tools makes you productive
 - Choosing the wrong ones will make your life tedious
- You do not need to know every tool by heart but...
 - ...you should know the landscape

Why 2026 is special

- Coding Agents like Claude Code became frightfully good this year
- Developing software is now less about writing *for loops* and *if else statements* yourself
- The less code you write yourself, the more important it is to understand the bigger picture
- AI can't compensate for choosing the wrong tool



Metaphorically speaking:

If you tell Claude to hammer a nail with a screwdriver into a wall, it will try to do so.

Who am I?

Marko Duda

- Masters in Cognitive Science with emphasis on AI and data science
- Co-published a deep learning paper
- Did research and engineering for Music Information Retrieval (at a start-up)
- Now a Natural Language Processing Engineer (at a mid-sized company)
- Wished that I had learned more about practical tools in university myself
- Python is my only programming language

Structure of this seminar

1	Friday, September 11, 2026	Quarto/ RStudio/ APA
2	Friday, September 18, 2026	Quarto/ RStudio/ APA
3	Friday, September 25, 2026	Introduction and first hands-on example
4	Friday, October 2, 2026	Operating Systems, basic security and CLIs
5	Friday, October 9, 2026	AI ecosystem: LLMs, tokens, agents, harnesses and more
6	Friday, October 16, 2026	Using Git, Github and Claude Code for Creating a Website
7	Friday, October 23, 2026	Docker and containerization
	— — —MID TERMS— — —	After the Mid Terms it is your turn
8	Friday, November 6, 2026	
9	Friday, November 13, 2026	
10	Friday, November 20, 2026	
11	Friday, November 27, 2026	
12	Friday, December 4, 2026	
13	Friday, December 11, 2026	
14	Friday, December 18, 2026	

Assessment methods and grading criteria

- PRESENTATION
 - 10–15 minute presentation
 - Recommend you doing live coding sessions
 - 3–5 page handout
 - Individual or group work
 - Maximum 5 students per group
 - Individual contributions must be clearly identifiable and assessable
- GRADING
 - Presentation: 80%
 - Handout: 20%
 - Pass: overall grade ≥ 4.0

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Content of this Seminar is on Github



- All content (slides etc) is uploaded to Github
 - <https://github.com/markoduda/Software-Applications-for-Computer-Science-WiSe-26-27>
 - Link is on ILIAS
- Github is a platform to develop software collaboratively
- Would advise you to create your own account
 - Great for showing future employees your coding project portfolio
 - Create your own Website for free

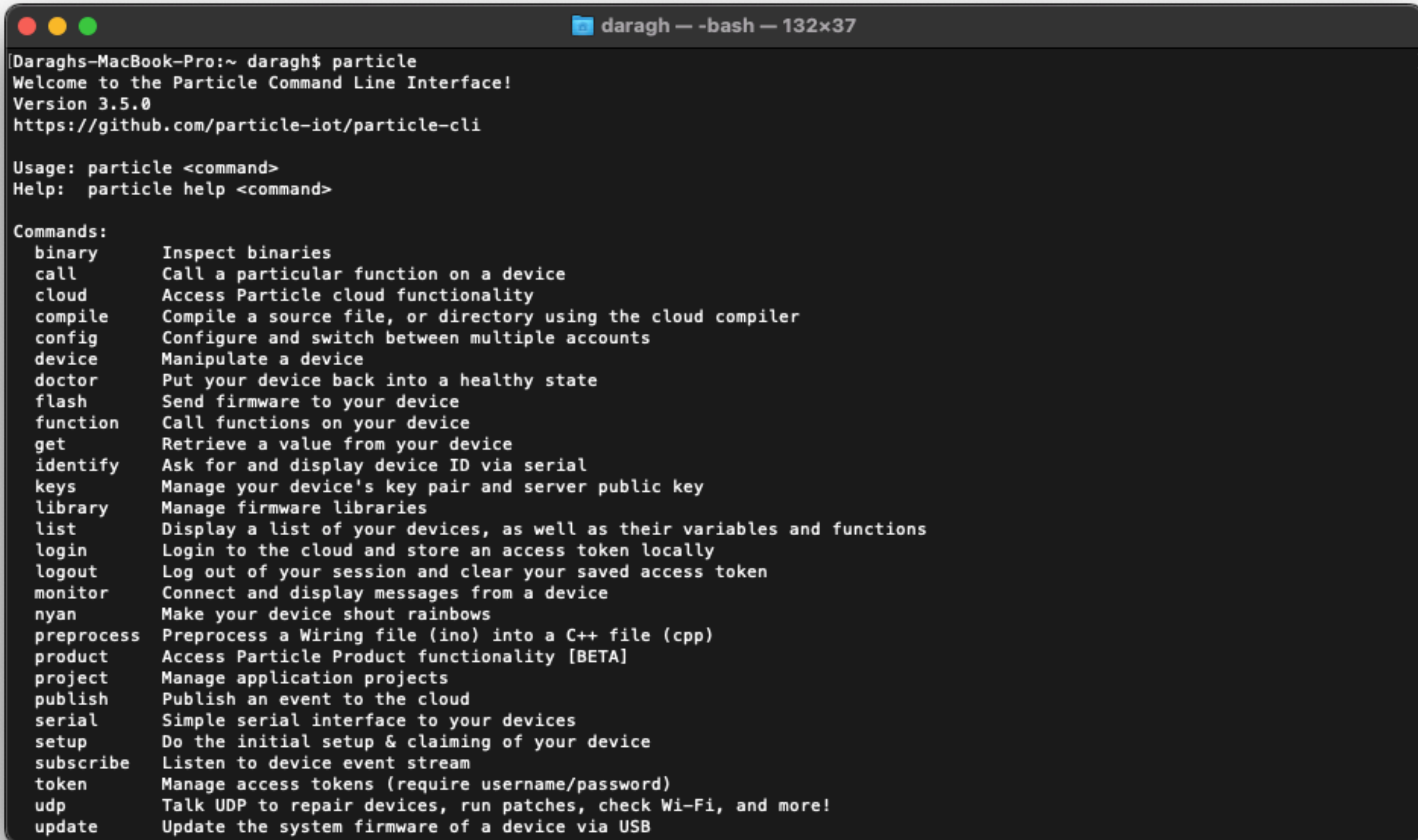
Who knows?

- IDE
- VS Code
- HTML
- DevOps
- Debugger
- Git
- GitHub
- GitLab
- Branch
- Merge
- Commit
- Pull/Merge request
- Version control
- Repository
- LaTeX
- Markdown
- Docker
- Containerization
- Kubernetes
- Unix
- Linux
- Windows Subsystem Linux
- Claude Code
- Codex
- Cursor
- OpenSource
- Shell scripts
- Command Line Interface
- Software as a service
- AWS
- Deployment
- README

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- Shell scripts
- **Command Line Interface**
- Software as a service
- AWS
- Deployment
- README

Command Line Interface (CLI)

A terminal window titled 'daragh — -bash — 132x37' with standard macOS window controls (red, yellow, green buttons). The terminal shows the command 'particle' being executed, which displays a welcome message, version information (3.5.0), and a link to the GitHub repository. It then lists usage and help information, followed by a comprehensive list of commands and their descriptions.

```
[Daraghs-MacBook-Pro:~ daragh$ particle]
Welcome to the Particle Command Line Interface!
Version 3.5.0
https://github.com/particle-iot/particle-cli

Usage: particle <command>
Help:  particle help <command>

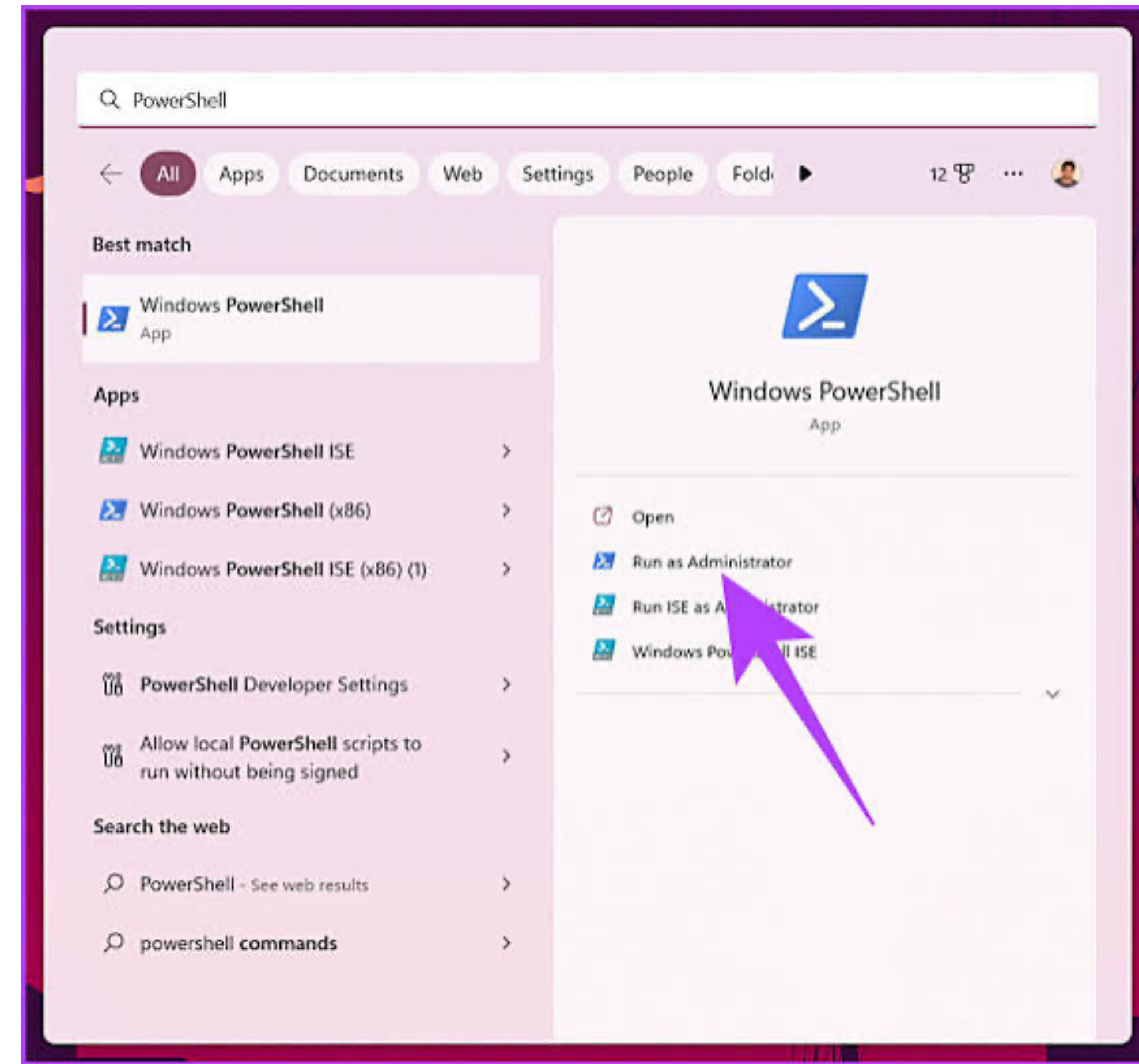
Commands:
binary      Inspect binaries
call        Call a particular function on a device
cloud       Access Particle cloud functionality
compile     Compile a source file, or directory using the cloud compiler
config      Configure and switch between multiple accounts
device      Manipulate a device
doctor      Put your device back into a healthy state
flash       Send firmware to your device
function    Call functions on your device
get         Retrieve a value from your device
identify    Ask for and display device ID via serial
keys        Manage your device's key pair and server public key
library     Manage firmware libraries
list        Display a list of your devices, as well as their variables and functions
login       Login to the cloud and store an access token locally
logout      Log out of your session and clear your saved access token
monitor     Connect and display messages from a device
nyan        Make your device shout rainbows
preprocess  Preprocess a Wiring file (ino) into a C++ file (cpp)
product     Access Particle Product functionality [BETA]
project     Manage application projects
publish     Publish an event to the cloud
serial      Simple serial interface to your devices
setup       Do the initial setup & claiming of your device
subscribe   Listen to device event stream
token       Manage access tokens (require username/password)
udp         Talk UDP to repair devices, run patches, check Wi-Fi, and more!
update      Update the system firmware of a device via USB
```

Command Line Interface (CLI)

- Offers faster speed, precise control and powerful automation that graphical UIs cannot match
- Many developer tools like *Docker* or *Git* work best with the CLI only
- Easy to use on remote computers
- AI and CLIs are a natural match (both work with text only)
- **We will use the Unix Command Line**
 - Nativ on Linux and Mac
 - Needs Windows Subsystem Linux for Win 11 (hope nobody uses Win 10 anymore)

Windows Subsystem Linux (WSL)

- WSL is a Linux virtual machine natively supported by Windows
- Easy setup
 - Open PowerShell in administrator mode by right-clicking and selecting "Run as administrator"
 - Enter the *wsl --install* command
 - Restart your machine
 - [Detailed Setup if really necessary](#)



Use Change Detection for Alerts on new Shared Flats

WG-GESUCHT

Angebote/Gesuche

Ratgeber & Services

Für Unternehmen

Bonitätscheck

Anzeige inserieren

Mein Konto

Köln

WG-Zimmer¹

unbefristet, befristet,³

Größe ▾

Miete ▾

MEINE FAVORITEN

Anzeigen auf Liste⁰

Leeren

Speichern

SUCHAUFTRAG EINRICHTEN

Erhalten Sie passende Angebote zu Ihrer Suche in Köln bequem per E-Mail.

Neue Angebote per E-Mail

ANZEIGENSUCHE

WG-Zimmer in Köln

606 WG-Zimmer | 994 Angebote in allen Kategorien in Köln

Wohnungen (255)

1-Zimmer-Wohnungen (125)

WG-Zimmer (606)

Häuser (8)

EINTRAGSDATUM ▾

Privatzimmer in Studenten-WG mittel- oder langfristig zu ...

Mitwohnmöglichkeit auf Zeit

ab 430 €

auch monatsweise

Beispielanzeige von HousingAnywhere*

Anzeige

Practical example:

Change detection for website monitoring

- 3 Options for Change Detection
- changedetection.io is about 9 Dollars a month
- Cheapest server is about 4.50 Euro a month
 - Code change detection yourself
 - **Use a ready-made docker container for change detection**

What I will do

Change detection for website monitoring

- How to rent a server
- Setup a remote connection to the server
- Run a change detection Docker container
- Access the container over a the browser

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